

EXAM 7 – SPRING 2015

11. (1.5 points)

An insurer is doing bootstrapping using a GLM with a log-link function and an over-dispersed Poisson error function on the following dataset of incremental reported loss values (\$000).

<u>Year</u>	<u>Age 1</u>	<u>Age 2</u>	<u>Age 3</u>	<u>Age 4</u>	<u>Age 5</u>
1	50	100	0	0	0
2	250	450	200	50	0
3	300	500	250	-20	0
4	250	1,000	100	-50	
5	300	500	250		
6	350	400			
7	300				

a. (0.25 point)

Identify an issue with the data that would cause model failure if left unadjusted.

b. (0.5 point)

Identify two additional issues with the data that may impact modeling results.

c. (0.75 point)

For each of the three issues identified in part a. and part b. above, briefly describe how the data might be adjusted to improve modeling results.

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EXAM 7 SPRING 2015 SAMPLE ANSWERS AND EXAMINERS' REPORT

QUESTION 11	
TOTAL POINT VALUE: 1.5	LEARNING OBJECTIVE: A7: Derive predictive distributions using bootstrapping and simulation techniques. A8: Identify data issues and related model adjustments for reserving models. A9: Test assumptions underlying reserve models.
SAMPLE ANSWERS	
Part a: 0.25 point	
<i>Sample Answer 1</i> Incremental values in Age 4 is negative <i>Sample Answer 2</i> Sum of incremental values in Age 4 is negative	
Part b: 0.5 point	
<i>Sample Answers from Multiple Candidates</i> Year 4, Ages 2 & 3 have values that seems to be outliers There appears to be a large increase in exposures from Year 1 to Year 2 Year 1 is likely the first year so the data is very thin Year 1 has a different exposure level There seems to be missing data (i.e. zeros) for Year 1, starting Age 3 The triangle seems to be incomplete due to the missing data (i.e. zeros) in Year 1	
Part c: 0.75 point	
<i>Sample Answers for Negative Incremental Values</i> Add 50 to each of the values in the triangle, solve the GLM and subtract 50 from the modeled result; OR Add 20 to each of the values in the triangle, solve the GLM and subtract 20 from the modeled result; OR Add a positive number to each of the values to eliminate the negative values in the triangle, solve the GLM and subtract the positive number from the modeled result; OR Subtract a negative number to each of the values to eliminate the negative values in the triangle, solve the GLM and add the negative number from the modeled result <i>Sample Answers for Outliers in Year 4, Ages 2 & 3</i> Could treat it as missing and estimate it from surrounding values Exclude Year 4 from the age-to-age factors (Age 1-2 and/or Age 2-3) and/or residual calculations Exclude outliers from the triangle <i>Sample Answer for Increased Exposure from Year 1 to Year 2</i> If earned exposure data is available, divide the whole loss triangle by exposures, using pure premium development (or loss ratio development) instead of total loss development <i>Sample Answer for Thin/Missing data in Year 1</i>	

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The entire row (for Year 1) can be removed from the loss triangle
EXAMINERS' REPORT
Candidates were expected to identify the issues which would cause model failure and impact modeling results, as well as how to address the identified issues prior to modeling. In general the candidates scored well. Most of the lost credit was from part c.
Part a
Candidates were expected to identify the negative incremental values in the triangle, which the majority did. The most common error was to identify the zeros in the triangle.
Part b
Candidates were expected to identify two additional issues that may impact modeling results (although they would not cause model failure). These were the outlier in Year 4, Age 2 and the data inconsistency between Year 1 and other years, perhaps due to a change in exposure or missing data. Most candidates could identify at least one of the issues. Common errors included misinterpreting the zeros in Age 5 as missing data (rather than claim closure) and identifying the negative incremental value in Age 4 (as it should have been identified in part a instead).
Part c
Candidates were expected to suggest adjustments to the data to improve modeling results, to address the three issues identified in parts a and b. See the sample answers above for acceptable suggestions. Most candidates provided full-credit solutions for how to adjust the data for outliers (Year 4) and missing/thin data (Year 1). The most common error was related to the negative incremental values issue; candidates described adding a value to each entry of the triangle before modeling, but they failed to mention subtracting the value back out from the modeled results. This response was considered incomplete because model output produced this way would be biased.